

# DJ SINDH GOVT. SCIENCE COLLEGE, KARACHI

Preliminary Examination 2026

Chemistry-XII

Wednesday  
11-02-2026

Time allowed: 3 hours

(Science Pre-Engineering & Pre Medical Group)

Maximum Marks: 85

Time allowed: 20 minutes

**SECTION 'A' (MULTIPLE CHOICE QUESTIONS) – (M.C.Qs)** (Marks: 17)

- NOTE: i) This section consists of 17 part questions and all are to be answered. Each part question carries one mark.  
ii) Do not copy the part questions in your answer book. Write only the answer in full against the proper number of the question and its part.  
iii) The use of scientific calculator is allowed

Q 1 Choose the correct answer for each from the given options.

(i) The oxidation states of the elements of group VII A:

- \* -1 and +3      \* +1, +5 and +5      \* +1, +1, +3, +5 and +7      \* +3, +3 and +5

(ii) The metal is used in making jewelry and also serves as catalyst:

- \* Pt      \* Fe      \* Pb      \* Al

(iii) Lucas reagent is a mixture of:

- \* Zn and Hg      \* Zn and HCl      \*  $ZnCl_2$  and HCl      \* NaOH and CaO

(iv) The best oxidizing agent among halogen is:

- \*  $I_2$       \*  $Br_2$       \*  $Cl_2$       \*  $F_2$

(v) Which one of the following does not belong to alkaline earth metal?

- \* Be      \* Ba      \* Ni      \* Ca

(vi) This is not a nucleophile:

- \*  $NH_3$       \* OH<sup>-</sup>      \*  $CN^-$       \*  $NH_4^+$

(vii) Welding gas among the following is:

- \* Ethene      \* Ethane      \* Methane      \* Ethyne

(viii) The element forms peroxide with oxygen:

- \* Na      \* K      \* Li      \* Mg

(ix) Formic acid is naturally found in:

- \* Venom of ants      \* Bee sting      \* Vinegar      \* Butter

(x) Saponification is the formation of soap by the reaction fat and oil with:

- \* Sugar      \* HCl      \* An Alkali      \* Glycerol

(xi) The coordination number of Ni in the complex compound  $[Ni(en)_2(NO_2)_2]^{2+}$ :

- \* 2      \* 4      \* 6      \* 5

(xii) Welding gas among the following is:

- \* Methane      \* Ethane      \* Ethylene      \* Acetylene

(xiii) Benzene burns with smokey flame because of its:

- \* Inflammability      \* High carbon percentage      \* High resonance energy      \* Aromaticity

(xiv) Drugs that lower the body temperature to normal are known as:

- \* Antibiotics      \* Antipyretic      \* Antiallergic      \* Anti histamines

(xv) This element is used to build strong bones and teeth:

- \* Zn      \* Ca      \* Fe      \* Mg

(xvi) Ozone is produced in the stratosphere region due to photo chemical reaction of sun rays and:

- \* CFCs      \*  $NO_2$       \* NO      \*  $O_2$

(xvii) Infra-red spectroscopy is a technique use to determine:

- \* Functional group      \* Double and triple bonds      \* Mass to charge ratio      \* Conjugated system

# DJ SINDH GOVT. SCIENCE COLLEGE, KARACHI



Preliminary Examination 2026

Chemistry – XII

Time allowed: 2 hrs. and 40 min. (Science Pre-Engineering & Pre Medical Group)

Wednesday  
11-02-2026

Max Marks: 68

## SECTION 'B' (SHORT-ANSWER QUESTIONS) (Marks : 36)

Note: Answer any Nine part questions. Select Four part questions from Inorganic – General Chemistry and Five part questions from Organic Chemistry. All part questions carry equal marks.

### INORGANIC-GENERAL CHEMISTRY

(Q.2) (i) Define the flame test. Briefly explain why alkali and alkaline earth metals produce colored flames in this test. State the flame colors for any two alkali metals. OR Give reasons for any four of the following:

\* Boiling point of halogens increases down the group in the periodic table. \* Electronegativity decreases regularly from top to bottom in s-block elements. \* Alkali metals are good conductor of electricity. \* Fluorine is the strongest oxidizing agent. \* Multidentate ligands are known as chelating agents.

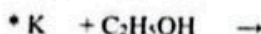
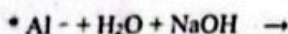
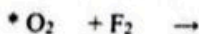
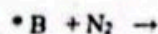
(ii) Write the diagonal relationship between lithium and magnesium. OR Write the three properties of beryllium that show its unique behavior in group II A. Also write the reason why beryllium does not react with cold water and steam?

(iii) Define d-block elements, why do they form colored compounds? Explain it in term of Crystal Field Theory.

(iv) Write IUPAC names of the following complexes:

\*  $[Cu(NH_3)_4]SO_4$  \*  $[Fe(CN)_5NO]^{2-}$  \*  $K_3[Cr(C_2O_4)_2Cl_2]$  \*  $[Pt(en)_2(NO_2)_2]Cl_2$

(v) Describe the preparation and two properties of Nylon and Polyvinyl chloride. OR Complete the following reactions.



(vi) Define green house gases. How green house gases cause to global warming? OR What are carbohydrates? Give their classification on the basis of structure.

(vii) What information about the structure of a molecule we can get from mass spectroscopy? Give the applications of mass spectroscopy.

### ORGANIC CHEMISTRY

(viii) Define Bucky ball. Explain its structure and write two properties. OR Define chiral carbon and plane polarized light. Explain optical isomerism briefly. OR Define Isomerism. Explain different types of Structural isomerism with examples.

(ix) Two organic compounds A and B are given below:

(A)  $CH_2=CH_2$

(B)  $CH \equiv CH$

\* Draw the orbital structure of compound A and specify: \* Hybridization \* Bond angles \* Write the ozonolysis reactions of A and B under suitable conditions. OR Give the equations and write the name of the final product in the following process.

\* Ethyne is treated with hydrogen bromide. \* Ethene is treated with oxygen in presence of peracetic acid at  $100^\circ C$ .

\* 1, 2-dibromoethane is heated with alcoholic KOH. \* Ethyne is treated with  $H_2O$  in the presence of  $H_2SO_4/HgSO_4$ .

(x) Define organometallic compounds. Give two examples of organo metallic compounds. Write the equation of methyl magnesium iodide with: \*  $CH_3CHO$  \*  $CO_2$

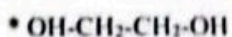
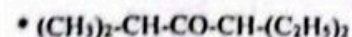
(xi) Why are amines basic in nature? How can we prepare ethyl amine from the following compounds? \* ethyl iodide \* methyl cyanide \* ethanamide OR What is photochemical reaction? Write the mechanism of chlorination of methane.

(xii) Name four derivative of carboxylic acid and write the equations of their preparation. OR Define Homologous series and write its three general properties.

(xiii) Identify each of the following with one laboratory test: \* Alcohol \* Phenol \* Alkene \* Aldehyde

OR Explain electrophilic substitution reaction mechanism in benzene with reference to halogenations.

(xiv) Write the IUPAC names of any four of the following:



**SECTION 'C' (DETAILED-ANSWER QUESTIONS) (Marks : 12)**

**Note:** Answer any **Four** questions. Two question from Inorganic – General Chemistry and Two from Organic Chemistry. All questions carry equal marks.

**INORGANIC-GENERAL CHEMISTRY**

**Q.3.** Explain with the help of diagram of Castner Kellner cell, how caustic soda is obtained by electrolysis of aqueous sodium chloride. Write two advantages and disadvantages of the Castner Kellner process. **OR** Describe the manufacturing of  $H_2SO_4$  by contact process. Show that by chemical equation it acts as oxidizing and dehydrating agent.

**Q.4.** Describe various steps involve in the extraction of 99.99% pure copper from its chalcopyrite ore. (Draw diagram where necessary) **OR** Write the balance chemical equations for the following:

- \* A mixture of carbon and silicon is heated under elevated temperature. \* Phosphorus is put into water.
- \* Bleaching powder is treated with hydrochloric acid. \* Chlorine gas is passed through hot aqueous solution of caustic soda. \* Copper is treated with concentration nitric acid. \* A piece of chromium is put into dilute hydrochloric acid.
- \* Reaction between  $KMnO_4$  and  $FeSO_4$  in the presence of  $H_2SO_4$  (write ionic equation)
- \* Reaction between  $K_2Cr_2O_7$  and  $FeSO_4$  in the presence of  $H_2SO_4$  (write ionic equation)

**Q.5.** Define Troposphere and stratosphere. Describe the chemistry involved due to the presence of oxides of carbon and nitrogen in the troposphere. (Write equations where necessary)

**ORGANIC CHEMISTRY**

**Q.6.** What is nucleophilic substitution reaction? Explain  $SN^1$  and  $SN^2$  mechanism with examples. **OR** Define  $\beta$  – elimination reaction. Discuss the mechanism of  $E_1$  and  $E_2$  reactions with examples. **OR** Define protein. Discuss types of protein on the basis of their function (any three). Write two properties and two biological importance of protein.

**Q.7.** Draw and explain the molecular orbital structure of Benzene. **OR** What is meant by orientation of benzene? Explain ortho-para and meta directing group. Write the equation for the preparation of TNT and m-nitro toluene from benzene.

**Q.8.** Draw the structure of the following organic molecules:

- |                       |                          |                    |                           |
|-----------------------|--------------------------|--------------------|---------------------------|
| * Isopropyl butanoate | * Ethyl neo-pentyl ether | * Divinylacetylene | * p-cresol                |
| * Pyrogallol          | * Neo Pentyl iodide      | * Benzamide        | * Benzene-1, 4-dioic acid |

**OR** Write the equations for the following reactions:

- \* Oxidation of  $2^\circ$  -alcohol with  $K_2Cr_2O_7/H_2SO_4$
- \* Reaction of phenol with  $H_2SO_4$  at  $20^\circ C$
- \* Reaction of propanone with hydroxyl amine.
- \* Ethanol, in excess, is heated up to  $140^\circ C$  in presence of  $H_2SO_4$
- \* Dehydration of ethyl alcohol at  $170^\circ C$  in conc.  $H_2SO_4$
- \* Reaction of ethylene glycol with periodic acid.
- \* Reduction of acetone with  $LiAlH_4$
- \* Oxidation of  $1^\circ$  -alcohol with PCC.

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**Good Luck**  
And Successful Results